

Assignment 4 – (Part1)

Mathematical Expectations

1. From an urn containing 3 red balls and 2 white balls, a man is to draw 2 balls at random without replacement. He gets rupees 20 for each red ball and rupees 10 for each white ball he draws. Find his expectation.(Ans: rs. 32)
2. A man with n keys in his pocket wants to open the door of his case by trying the keys independently and randomly one by one. Find the mean and variance of the number of trials required to open the door if unsuccessful keys are kept aside. (Ans: $(n+1)/2, (n^2-1)/12$)
3. In a game of chance a man is allowed to throw a fair coin indefinitely. He receives rupees 1,2,3.....if he throws a head at the 1st, 2nd, 3rd trail respectively. If the entry fee to participate in the game is rupees 2, find the expected value of his net gain.(Ans: zero)
4. Three fair coins are tossed. Find the expectation and the variance of number of heads. (Ans: 3/2, 3/4)
5. A continuous random variable X has the density function $f(x) = k(1+x)$ where $2 \leq x \leq 5$. Find k , $P(x \leq 4)$ and $E(X)$.(Ans: 2/27, 16/27, 11/3)
6. Find the mean and variance of the following distribution (Ans:3, 3)

X	1	3	4	5
P(X=x)	0.4	0.1	0.2	0.3

7. If probability density of a random variable is given by

$$f(x) = \begin{cases} kx, & 0 \leq x \leq 2 \\ 2k, & 2 \leq x \leq 4 \\ 6k - kx, & 4 \leq x \leq 6 \end{cases}$$

Find (i) k , (ii) $P(1 \leq X \leq 3)$, (iii) $\text{mean}(\bar{x})$. (Ans: 1/8, 7/16, 7/4)

8. A continuous random variable has probability density function

$$f(x) = \begin{cases} 2e^{-2x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

Find (i) $E(X)$, (ii) $E(X^2)$, (iii) $\text{Var}(X)$, (iv) S.D of X . (Ans: 1/2, 1/2, 1/4, 1/2)

9. A continuous random variable X has the probability density function given by

$$f(x) = \begin{cases} 2ax + b, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

If the mean of the distribution is 3, find the constants a and b .(Ans: 3/2, -5/2)